```
==================================================
-- ============================================================
-- INTERMEDIATE EXERCISES
-- Topics: JOINs, GROUP BY, HAVING, Aggregates, Subqueries
-- ============================================================

-- Exercise 1
-- Show each employee's full name along with their department name.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    d.name AS department_name
FROM employees e
JOIN departments d
    ON e.department_id = d.id
```

employee_name varchar	department_name varchar
Sam CEO	Engineering
Alice Smith	Engineering
David Brown	Sales
Grace Wilson	HR
Iris Taylor	Marketing
Karen Thomas	Finance
Bob Jones	Engineering
Eva Davis	Sales
Leo Jackson	Finance
Carol White	Engineering
.	.
.	.
.	.
Henry Moore	HR
Rachel Scott	HR
Steve King	HR
Jack Anderson	Marketing
Tina Wright	Marketing
Uma Lopez	Marketing
Victor Hill	Finance
Wendy Green	Finance
Xander Baker	Engineering
Yara Nelson	Engineering

26 rows (20 shown)   2 columns

```sql
==================================================
-- Exercise 2
-- Show each employee's full name and their manager's full name.
-- If an employee has no manager, still show their name (with NULL for manager).
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    CONCAT(m.first_name, ' ', m.last_name) AS manager_name
FROM employees e
LEFT JOIN employees m
    ON e.manager_id = m.id
```

employee_name varchar	manager_name varchar
Alice Smith	Sam CEO
David Brown	Sam CEO
Grace Wilson	Sam CEO
Iris Taylor	Sam CEO
Karen Thomas	Sam CEO
Bob Jones	Alice Smith
Eva Davis	David Brown
Leo Jackson	Karen Thomas
Carol White	Bob Jones
Mia Harris	Bob Jones
.	.
.	.
.	.
Rachel Scott	Grace Wilson
Steve King	Grace Wilson
Jack Anderson	Iris Taylor
Tina Wright	Iris Taylor
Uma Lopez	Iris Taylor
Victor Hill	Leo Jackson
Wendy Green	Leo Jackson
Xander Baker	Bob Jones
Yara Nelson	Bob Jones
Sam CEO	

26 rows (20 shown) 2 columns

```
=================================================
-- Exercise 3
-- Find the total number of employees in each department.
-- Show department name and employee count, sorted by count descending.
SELECT
    d.name AS department_name,
    COUNT(e.id) AS employee_count
FROM departments d
LEFT JOIN employees e
    ON d.id = e.department_id
GROUP BY d.id, d.name
ORDER BY employee_count DESC
```

department_name varchar	employee_count int64
Engineering	8
Sales	6
HR	4
Marketing	4
Finance	4

```
=================================================
-- Exercise 4
-- Find the average salary per department.
-- Only show departments where the average salary is above 80,000.
SELECT
    d.name AS department_name,
    AVG(e.salary) AS average_salary
FROM departments d
JOIN employees e
    ON d.id = e.department_id
GROUP BY d.id, d.name
HAVING AVG(e.salary) > 80000
ORDER BY average_salary DESC
```

department_name varchar	average_salary double
Engineering	104500.0
Finance	86000.0
Marketing	81750.0

```
==================================================
-- Exercise 5
-- Find the highest and lowest salary in the company per job title.
SELECT
    job_title,
    MAX(salary) AS highest_salary,
    MIN(salary) AS lowest_salary
FROM employees
GROUP BY job_title
ORDER BY job_title
```

job_title varchar	highest_salary decimal(10,2)	lowest_salary decimal(10,2)
CEO	200000.00	200000.00
CFO	115000.00	115000.00
Engineer	88000.00	82000.00
Finance Manager	80000.00	80000.00
Financial Analyst	76000.00	73000.00
HR Analyst	67000.00	67000.00
HR Director	95000.00	95000.00
HR Specialist	65000.00	63000.00
Junior Engineer	79000.00	79000.00
Marketing Analyst	74000.00	70000.00
Marketing Director	105000.00	105000.00
Sales Manager	75000.00	75000.00
Sales Rep	72000.00	68000.00
Senior Engineer	92000.00	90000.00
Senior Marketing Analyst	78000.00	78000.00
VP Engineering	120000.00	120000.00
VP Sales	110000.00	110000.00

17 rows 3 columns

```
==================================================
-- Exercise 6
-- Show each project's name and the number of employees assigned to it.
-- Include projects that have no employees assigned.
SELECT
    p.name AS project_name,
    COUNT(ep.employee_id) AS employee_count
FROM projects p
LEFT JOIN employee_projects ep
    ON p.id = ep.project_id
GROUP BY p.id, p.name
ORDER BY p.name
```

project_name varchar	employee_count int64
Analytics Dashboard	7
CRM Integration	5
Data Migration	3
Finance Reporting	4
HR Portal	3
Mobile App v2	6
Security Audit	3
Website Redesign	3

```sql
-- Exercise 7
-- Find all employees who are assigned to more than one project.
-- Show their name and the number of projects.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    COUNT(ep.project_id) AS project_count
FROM employees e
JOIN employee_projects ep
    ON e.id = ep.employee_id
GROUP BY e.id, e.first_name, e.last_name
HAVING COUNT(ep.project_id) > 1
ORDER BY project_count DESC, employee_name
```

employee_name varchar	project_count int64
Carol White	3
Olivia Garcia	3
Victor Hill	3
Bob Jones	2
Grace Wilson	2
Henry Moore	2
Leo Jackson	2
Wendy Green	2

```sql
-- Exercise 8
-- Find the total sales amount per sales employee.
-- Show their full name and total sales, sorted by total sales descending.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    SUM(s.amount) AS total_sales_amount
FROM employees e
JOIN sales s
    ON e.id = s.employee_id
GROUP BY e.id, e.first_name, e.last_name
ORDER BY total_sales_amount DESC
```

employee_name varchar	total_sales_amount decimal(38,2)
Nathan Martin	146000.00
Eva Davis	137500.00
Frank Miller	128500.00
Paula Lee	96000.00
Quinn Adams	75500.00

```sql
-- Exercise 9
-- Find all employees who earn more than the average salary of their department.
-- Show their name, salary, and department name.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    e.salary,
    d.name AS department_name
FROM employees e
JOIN departments d
    ON e.department_id = d.id
WHERE e.salary > (
    SELECT AVG(e2.salary)
    FROM employees e2
    WHERE e2.department_id = e.department_id
)
ORDER BY e.salary DESC
```

employee_name varchar	salary decimal(10,2)	department_name varchar
Sam CEO	200000.00	Engineering
Alice Smith	120000.00	Engineering
Karen Thomas	115000.00	Finance
David Brown	110000.00	Sales
Iris Taylor	105000.00	Marketing
Grace Wilson	95000.00	HR

```sql
-- Exercise 10
-- List all employees who have NOT been assigned to any project.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name
FROM employees e
LEFT JOIN employee_projects ep
    ON e.id = ep.employee_id
WHERE ep.employee_id IS NULL
ORDER BY employee_name
```

employee_name varchar
Nathan Martin
Quinn Adams
Sam CEO

```sql
-- Exercise 11
-- For each department, show the name of the highest-paid employee.
SELECT
    d.name AS department_name,
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name
FROM departments d
JOIN employees e
    ON d.id = e.department_id
WHERE e.salary = (
    SELECT MAX(e2.salary)
    FROM employees e2
    WHERE e2.department_id = d.id
)
ORDER BY d.name
```

department_name varchar	employee_name varchar
Engineering	Sam CEO
Finance	Karen Thomas
HR	Grace Wilson
Marketing	Iris Taylor
Sales	David Brown

```sql
-- Exercise 12
-- Find all projects where the total hours logged by all employees exceeds 400.
-- Show project name and total hours.
SELECT
    p.name AS project_name,
    SUM(ep.hours_logged) AS total_hours
FROM projects p
JOIN employee_projects ep
    ON p.id = ep.project_id
GROUP BY p.id, p.name
HAVING SUM(ep.hours_logged) > 400
ORDER BY total_hours DESC
```

project_name varchar	total_hours int128
Mobile App v2	1485
Analytics Dashboard	1310
Data Migration	580
Website Redesign	540
CRM Integration	505
Finance Reporting	500

```sql
-- Exercise 13
-- Show each employee's full name, their department name, and their manager's full name.
-- If no manager, show 'No Manager'.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    d.name AS department_name,
    COALESCE(CONCAT(m.first_name, ' ', m.last_name), 'No Manager') AS manager_name
FROM employees e
JOIN departments d
    ON e.department_id = d.id
LEFT JOIN employees m
    ON e.manager_id = m.id
```

employee_name varchar	department_name varchar	manager_name varchar
Alice Smith	Engineering	Sam CEO
David Brown	Sales	Sam CEO
Grace Wilson	HR	Sam CEO
Iris Taylor	Marketing	Sam CEO
Karen Thomas	Finance	Sam CEO
Bob Jones	Engineering	Alice Smith
Eva Davis	Sales	David Brown
Leo Jackson	Finance	Karen Thomas
Carol White	Engineering	Bob Jones
Mia Harris	Engineering	Bob Jones
·	·	·
·	·	·
·	·	·
Rachel Scott	HR	Grace Wilson
Steve King	HR	Grace Wilson
Jack Anderson	Marketing	Iris Taylor
Tina Wright	Marketing	Iris Taylor
Uma Lopez	Marketing	Iris Taylor
Victor Hill	Finance	Leo Jackson
Wendy Green	Finance	Leo Jackson
Xander Baker	Engineering	Bob Jones
Yara Nelson	Engineering	Bob Jones
Sam CEO	Engineering	

26 rows (20 shown) 3 columns

```sql
-- Exercise 14
-- Find the total sales per product.
-- Show product name and total amount, sorted by total amount descending.
SELECT
    s.product,
    SUM(s.amount) AS total_amount
FROM sales s
GROUP BY s.product
ORDER BY total_amount DESC
```

product varchar	total_amount decimal(38,2)
Enterprise	276000.00
Pro Plan	243000.00
Starter Plan	64500.00

```sql
-- Exercise 15
-- Find all employees who share the same job title.
-- Show the job title and the names of employees who have it.
-- Exclude job titles held by only one person.
SELECT
    job_title,
    CONCAT(first_name, ' ', last_name) AS employee_name
FROM employees
WHERE job_title IN (
    SELECT job_title
    FROM employees
    GROUP BY job_title
    HAVING COUNT(*) > 1
)
ORDER BY job_title, employee_name
```

job_title varchar	employee_name varchar
Engineer	Carol White
Engineer	Mia Harris
Engineer	Xander Baker
Financial Analyst	Victor Hill
Financial Analyst	Wendy Green
HR Specialist	Henry Moore
HR Specialist	Rachel Scott
Marketing Analyst	Jack Anderson
Marketing Analyst	Tina Wright
Sales Rep	Frank Miller
Sales Rep	Nathan Martin
Sales Rep	Paula Lee
Sales Rep	Quinn Adams
Senior Engineer	Bob Jones
Senior Engineer	Olivia Garcia

15 rows 2 columns

```sql
-- Exercise 16
-- For each department, show the number of employees hired after 2020.
SELECT
    d.name AS department_name,
    COUNT(e.id) AS employee_count
FROM departments d
JOIN employees e
    ON d.id = e.department_id
WHERE e.hire_date > '2020-12-31'
GROUP BY d.id, d.name
ORDER BY employee_count DESC
```

department_name varchar	employee_count int64
Sales	5
HR	2
Finance	2
Engineering	2
Marketing	1

```sql
==================================================
-- Exercise 17
-- Find the employee who has logged the most total hours across all projects.
-- Show their name and total hours.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    SUM(ep.hours_logged) AS total_hours
FROM employees e
JOIN employee_projects ep
    ON e.id = ep.employee_id
GROUP BY e.id, e.first_name, e.last_name
ORDER BY total_hours DESC
LIMIT 1
```

employee_name varchar	total_hours int128
Carol White	690

```sql
-- Exercise 18
-- Show each region's total sales and the number of sales transactions.
SELECT
    region,
    SUM(amount) AS total_sales,
    COUNT(*) AS sales_transactions
FROM sales
GROUP BY region
ORDER BY total_sales DESC
```

region varchar	total_sales decimal(38,2)	sales_transactions int64
East	359000.00	20
West	224500.00	14

```sql
-- Exercise 19
-- Find all employees who are both a manager (someone reports to them)
-- and are assigned to at least one project.
-- Show their name, department, and number of direct reports.
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    d.name AS department_name,
    COUNT(DISTINCT r.id) AS direct_reports
FROM employees e
JOIN departments d
    ON e.department_id = d.id
JOIN employee_projects ep
    ON e.id = ep.employee_id
JOIN employees r
    ON r.manager_id = e.id
GROUP BY e.id, e.first_name, e.last_name, d.name
ORDER BY direct_reports DESC, employee_name
```

employee_name varchar	department_name varchar	direct_reports int64
Bob Jones	Engineering	5
Eva Davis	Sales	4
Grace Wilson	HR	3
Iris Taylor	Marketing	3
Leo Jackson	Finance	2
Alice Smith	Engineering	1
David Brown	Sales	1
Karen Thomas	Finance	1

```sql
====================================================
-- Exercise 20
-- List each project with its total budget vs total hours logged.
-- Show: project name, budget, total_hours, and cost_per_hour
-- (assume cost_per_hour = budget / total_hours, rounded to 2 decimal places).
SELECT
    p.name AS project_name,
    p.budget,
    SUM(ep.hours_logged) AS total_hours,
    CASE
        WHEN SUM(ep.hours_logged) = 0 THEN NULL
        ELSE ROUND(p.budget / SUM(ep.hours_logged), 2)
    END AS cost_per_hour
FROM projects p
LEFT JOIN employee_projects ep
    ON p.id = ep.project_id
GROUP BY p.id, p.name, p.budget
ORDER BY p.name
```

project_name varchar	budget decimal(12,2)	total_hours int128	cost_per_hour double
Analytics Dashboard	70000.00	1310	53.44
CRM Integration	95000.00	505	188.12
Data Migration	80000.00	580	137.93
Finance Reporting	60000.00	500	120.0
HR Portal	45000.00	330	136.36
Mobile App v2	120000.00	1485	80.81
Security Audit	30000.00	105	285.71
Website Redesign	50000.00	540	92.59

```
================================================================
-- Exercise 21
-- Find the top-selling product in each region.
-- Show: region, product, and total sales amount.
SELECT
    region,
    product,
    total_sales_amount
FROM (
    SELECT
        s.region,
        s.product,
        SUM(s.amount) AS total_sales_amount,
        ROW_NUMBER() OVER (
            PARTITION BY s.region
            ORDER BY SUM(s.amount) DESC
        ) AS rn
    FROM sales s
    GROUP BY s.region, s.product
) ranked
WHERE rn = 1
ORDER BY region
```

region varchar	product varchar	total_sales_amount decimal(38,2)
East	Enterprise	202000.00
West	Pro Plan	132000.00

```sql
-- Show all employees who joined in the same year as at least one other employee
-- from a different department.
-- Show their name, department name, and hire year.
SELECT
    CONCAT(e1.first_name, ' ', e1.last_name) AS employee_name,
    d.name AS department_name,
    EXTRACT(YEAR FROM e1.hire_date) AS hire_year
FROM employees e1
JOIN departments d
    ON e1.department_id = d.id
WHERE EXISTS (
    SELECT 1
    FROM employees e2
    WHERE EXTRACT(YEAR FROM e2.hire_date) = EXTRACT(YEAR FROM e1.hire_date)
      AND e2.department_id <> e1.department_id
)
ORDER BY hire_year, employee_name
```

employee_name varchar	department_name varchar	hire_year int64
Alice Smith	Engineering	2018
Olivia Garcia	Engineering	2018
Uma Lopez	Marketing	2018
Bob Jones	Engineering	2019
Iris Taylor	Marketing	2019
Steve King	HR	2019
Carol White	Engineering	2020
Leo Jackson	Finance	2020
Mia Harris	Engineering	2020
Tina Wright	Marketing	2020
Eva Davis	Sales	2021
Frank Miller	Sales	2021
Rachel Scott	HR	2021
Victor Hill	Finance	2021
Xander Baker	Engineering	2021
Henry Moore	HR	2022
Nathan Martin	Sales	2022
Paula Lee	Sales	2022
Wendy Green	Finance	2022
Jack Anderson	Marketing	2023
Quinn Adams	Sales	2023
Yara Nelson	Engineering	2023